Deep Learning & Neural Network Architectures

Department: AI & Data Science | Semester: 7 | Code: AD702
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1. Multilayer Perceptrons & Backpropagation

Forward pass computes activations layer by layer: z^[l] = W^[l]*a^[l-1] + b^[l]. Backpropagation uses the chain rule to compute gradients of the loss with respect to weights.

2. Convolutional Neural Networks (CNNs)

Convolutional layers apply spatial filters to extract translation-invariant visual features, followed by ReLU activations and max-pooling.

3. Recurrent Architectures & Transformers

LSTMs and GRUs address vanishing gradients in sequential data. Transformers use multi-head self-attention mechanisms to process sequence tokens in parallel.